

A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 06

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Area must be large enough to support viable populations (breeding numbers high enough, gene pool large enough to avoid inbreeding); species higher up food chains need larger areas (lower density) • Large areas contain a full range of resources (eg different fruiting trees / feeding areas through the year) • Shape: the perimeter has edge conditions (a mix of neighbouring habitats); a compact shape maximises core habitat for species needing core conditions • Some species benefit from edges (higher light) or small isolated habitats (ponds too small for predatory fish suit amphibians)	4
01.2	• Varied topography: banks, lagoons, islands, ditches created with machinery give a range of water depths (emergent, floating-leaved, submerged plant zones) • Islands provide roosting / nesting sites safe from terrestrial predators (foxes) • Control of salinity (eg intertidal conditions for opossum shrimps feeding dunlin and avocet); scrapes for skylarks; different water depths for waders and ducks	3
01.3	• Mobile species (birds, flying insects) colonise easily • Less mobile species (many plants, amphibians, invertebrates) may not reach an isolated site so must be introduced; keystone species especially	2
01.4	• Allows dispersal of surplus young, mixing of gene pools (less inbreeding) and recolonisation if a population dies out; avoids hazards (eg rope bridges for howler monkeys over roads)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 40-42; AQA spec 3.1.2.3.4.	

Question	Marking guidance	Marks
02.1	<i>Any 3 from:</i> • Sinkers / weighted lines so hooks sink quickly below the reach of albatrosses • Setting lines at night when birds are not feeding • Bird-scaring streamers / decoys • Curved hooks that catch tuna but not birds	3
02.2	• Effort: limits on boat size, engine power, days at sea, number of licences; decommissioning payments to reduce fleet size where overfishing is unavoidable • Methods: bans or length limits on drift nets; bans on demersal trawling over sensitive seabed (Rockall Bank, Hatton Bank deep-water reefs) • Shorter trawl tows so turtles / by-catch survive (over 99% survive under 10 minutes)	3
02.3	• Eggs / larvae are reared in captivity through the planktonic stage when most die (predation, failure to find habitat) • Juveniles are released when survival chances are good, increasing recruitment to the adult population	2
02.4	• Egg / larval survival varies greatly between years so adult numbers do not predict recruitment • Several poor years would lower the MSY; egg surveys allow catches to be reduced in advance rather than after a sudden collapse	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 344-348; AQA spec 3.5.2.2.	
03.1	• Giant tritons are predators of the crown-of-thorns starfish which eats coral polyps • Removing tritons allowed starfish populations to explode, destroying large areas of reef (food web / trophic cascade)	2
03.2	• Fixed mooring buoys so boats do not anchor on coral • Education of divers / snorkellers not to touch coral or kick up sediment; bans on spear fishing and souvenir collection • Fishing controls: no-take zones (Chagos, Great Barrier Reef, Lundy, Lamlash Bay), seasons, minimum and maximum catch sizes, bans on dynamite / small-mesh nets / trawling • Protection of turtle nesting beaches; control of coastal development; protection of mangroves and seagrass; artificial reefs (reef balls, sunk ships / oil rigs)	4
03.3	• No sunlight so no symbiotic algae; polyps rely only on captured food so have far less energy and grow very slowly • Damage from trawls takes hundreds of years to repair; the slow-growing, low-fecundity fish (orange roughy) they support are easily over-exploited	2
03.4	• Deep-water trawling (also by-catch); oil and gas exploration in deeper water; ocean acidification • Protected areas banning trawling, eg Darwin Mounds SAC off NW Scotland, Oculina Bank off Florida	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 61-64; AQA spec 3.1.2.3.4.	

Question	Marking guidance	Marks
04.1	• Cycling of materials (like biogeochemical cycles) - recycling, re-use • Energy from renewable sources • Human activities should support ecosystems / conserve biodiversity • Separation of technical and biological materials; connected systems where the waste of one process feeds another (industrial symbiosis) • Diverse systems are more resistant to change • Design of products for extended use and end-of-life re-use; optimum rather than maximum production; technologies that improve efficiency and enable dismantling	6
04.2	• Natural systems over-produce so the surplus supports the processes the system relies on: plant products feed pollinators, seed dispersers, decomposers, mycorrhizae • Agroecosystems aim for maximum harvested yield, removing the surplus that would maintain soil biota, pollinators and pest predators, so they need artificial inputs	2
04.3	• Costs of environmental measures could put companies in one country at a disadvantage against competitors in countries without them • A common approach gives a level playing field, coordinated monitoring and support for less successful countries	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 374-375; AQA spec 3.6.4.1.	
05.1	• Irrigation depletes rivers and aquifers (eg Aral Sea, Ogallala); increases evaporation • Soil erosion and compaction reduce water retention and infiltration so runoff and river flow fluctuations increase • Evapotranspiration changes: increased in irrigated arid areas, decreased where forests were cleared	3
05.2	• Seasonal / local: avoids energy for heating, lighting greenhouses out of season and long-distance transport (food miles) and the pollution this causes • Organic: uses natural processes (rotation, legumes, manure, biological control, predator habitats) instead of artificial fertilisers and pesticides, so less pollution, energy use and harm to non-target species; free-range and Fairtrade also accepted for welfare / social reasons	4
05.3	• Food shortages after 1945 (dependence on US food aid); farmers could not afford loans to invest so grants funded machinery, drainage, hedgerow removal, liming, improved livestock • Guaranteed market / price support: government bought surpluses to hold prices up (and sold stores in poor years) giving farmers security to increase output • Consequences: loss of hedgerows and wetlands, intensification, pesticide and fertiliser pollution; later food surpluses (butter mountains) that could not be sold without harming LEDC producers	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 329-333; AQA spec 3.5.1.3-3.5.1.4.	
06.1	• B	1

Question	Marking guidance	Marks
06.2	<i>Any 4 from:</i> • Harvesting no faster than the MSY - resource never depleted • Mixed species plantations - more food choices, relationships and niches • Indigenous species - support indigenous insects, birds • Mixed age structure - varied light, temperature, wind so more species; continuous cover and yield • Selective logging - small clearings, less disturbance, long-term timber benefit • Retention of dead wood, buffer zones, corridors	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-369; AQA spec 3.5.3.	
07.1	• Drainage of wetlands / lowering water tables by groundwater abstraction dries out wetlands; HEP schemes alter river levels flooding sandbank nests of turtles / birds; species with narrow tolerance (specialists) affected first; sundews lost when drained soil becomes aerobic / nitrogen-rich and taller competitors invade • Hot water discharges and organic wastes (sewage) reduce dissolved oxygen, killing aerobic organisms such as fish and insect larvae • Acid mine drainage and acid rain gases lower pH, denaturing proteins of exposed tissues (fish eggs, gills) and dissolving calcium exoskeletons (crayfish)	6
07.2	• Reduces light penetration so submerged plants cannot photosynthesise • Blocks the gills of filter feeders such as bivalve molluscs; sediment smothers the bed	2
07.3	• Rare species often have very specialised habitat requirements / narrow ranges of tolerance • A small change moves conditions outside their range while generalists survive; small populations cannot recover	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 22-23; AQA spec 3.1.2.2.	
08.1	<i>Any 4 from:</i> • Designation of protected areas (SSSIs, MCZs) • Protection of wild birds and their nests (except some pests / game birds) • Protection of named mammals such as otters, dormice, red squirrels, badgers and their setts • Uprooting wild plants generally illegal • Bats must not be disturbed; roof timber treatments must be non-toxic to bats	4
08.2	• Appendix I: species threatened with extinction - all international (commercial) trade banned, except movement for conservation breeding • Examples: great apes, big cats, rhinos, blue whale • Appendix II: species that may become threatened if trade is not closely controlled - trade permitted under licence from countries where the species is well protected • Examples: Honduras mahogany, hippopotamus, green iguana, great white / basking shark, Venus fly trap	4
08.3	• Collection may be hard to police in remote areas / poor countries, but if the animals or products cannot be exported / imported there is no market • Removing the profit removes the incentive to collect	2

Question	Marking guidance	Marks
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 30-31; AQA spec 3.1.2.3.2.	
09.1	• Negative • Positive • Negative • Positive	4
09.2	• The response to a change acts to re-establish / return the system to the previous equilibrium • So the change does not become self-sustaining unless a large unusual event overwhelms the negative feedback	2
09.3	• eg the possible release of methane hydrate at the end of the Permian era • Overwhelmed the negative feedback mechanisms, causing climate change and a mass extinction / a new equilibrium	2
09.4	• Soil degradation, reduced water availability and loss of natural pest predators would reduce yields (negative feedback curtailing the activity) • Instead of reducing the activity, farmers use fertilisers, irrigation and pesticides, which further damage the natural systems that support food production	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 371-372; AQA spec 3.6.1.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Resources: wood, fibres (cotton, wool, silk), oils, fuels (wood, biofuels), food; new food species (morama bean, perennial cereals, eland, bison) better adapted to local conditions • Biomimetics: vehicle design, architecture, structures, adhesion, materials, ultrasound • Medicines from plants and animals (examples); most species unstudied - argument for habitat conservation • Physiological research using wildlife species; wildlife as pest control agents (ladybirds, Encarsia, Cactoblastis, Bt) • Genetic resources: crop wild relatives, Vavilov centres, gene pools • Ecosystem services: atmospheric composition, hydrological and biogeochemical cycles, soil maintenance, pollination, seed dispersal, habitat provision • Ethical / aesthetic / recreational arguments; the unknown value of species not yet investigated (precautionary principle) • Evaluation: which arguments are most persuasive to decision makers; economic valuation vs intrinsic value; judgement	9
10.2	• The variety and abundance of life (species, genes, habitats) in an area / on Earth	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1.	

Question	Marking guidance	Marks
11.1	Discuss how an understanding of energy flow and the manipulation of food species can be used to increase food production sustainably. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Energy flow: autotrophs and heterotrophs; losses by reflection, transmission, respiration; 100 000 to 1 000 units; few food chains beyond four levels; plant food most productive; grazing animals where crops impossible; pigs on wastes • Energetics: intensive vs extensive, law of diminishing returns, energy subsidies (fertiliser, pesticides, irrigation, machinery, drying, processing, transport), energy ratios, food conversion ratios, dependence on fossil fuels • Control of genetics: asexual reproduction / cuttings / cloning, selective breeding and breeds, cross-breeding and hybrid vigour, AI and embryo transfer • GM: Roundup Ready soya, Bt crops, golden rice, omega-3 oilseed rape; advantages; risks (gene transfer, antibiotic markers, allergies, costs to LEDC farmers, loss of local diversity) • Sustainability links: energy dependence, trophic level of diet, feeding 12 billion people • Justified conclusion on the balance between productivity, efficiency and sustainability Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-328; AQA spec 3.5.1.2.	25

Question	Marking guidance	Marks
11.2	Using tropical coral reefs as an example, discuss how the ecological features of a habitat determine both its importance and its vulnerability to human activities. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Ecological features: cnidarian polyps, cilia, nematocysts, zooxanthellae symbiosis, limestone skeletons building reefs, complex structure and food web; specific abiotic requirements (temperature, light, salinity, low turbidity) • Importance: quarter of marine species, fisheries, medicines, carbon sink, erosion protection, tourism • Vulnerability: narrow abiotic tolerance means sensitivity to warming, acidification, sedimentation, nutrient pollution, oil, bleaching; slow growth so slow recovery; physical damage; overfishing and food-web effects (crown of thorns); introduced species; dependence on mangroves and seagrass • Conservation: MPAs, No Take Zones, fishing controls, tourism management, protection of linked habitats, CITES • Comparison with other habitats (rainforests, mangroves, Antarctica) where relevant • Evaluation of local vs global threats and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-62; AQA spec 3.1.2.3.4.	25